Polytropic hat-trick

Y20 (2020) — English translation

Once Lord Kelvin carried out a cycle of a heat engine, the working fluid of which was one mole of an ideal gas with adiabatic index γ. The cycle consists of three polytropic processes, one of which is isobaric, and the product of the exponents of the other two polytropes 13 and 23 was equal to $n_{13}n_{23} = -1$, where 1 and 2 are the isobar points with minimum and maximum temperatures, respectively, and 3 is the intersection point of the polytrope 13 and 23. The temperature at point 3 satisfies the relation $T_3^2 = T_1 T_2$. $T_1 = 100$ K, $T_2 = 400$ K, $\gamma = 1.5$.

A1	Find the work A of the gas in the cycle.	**3.50**

A2	Find the efficiency η of the cycle.	**1.50**

Source: https://pho.rs/p/28